

Unsupervised Learning for Steel Surface Defect Classification

A Re-implementation and Sensitivity Analysis of the Cohn & Holm
Method

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Outline

- 1 Introduction & Motivation
- 2 The Unsupervised Pipeline
- 3 Re-implementation Results
- 4 Key Sensitivity Analyses
- 5 Conclusion

The Challenge in Materials Science

- Microstructure analysis is fundamental to understanding material properties.

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 - **Not Scalable:** A bottleneck for high-throughput experiments.

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- Traditional analysis is:
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 - **Subjective:** Requires expert knowledge and can vary between analysts.
 - **Not Scalable:** A bottleneck for high-throughput experiments.
- **Goal:** Automate the classification of material images.

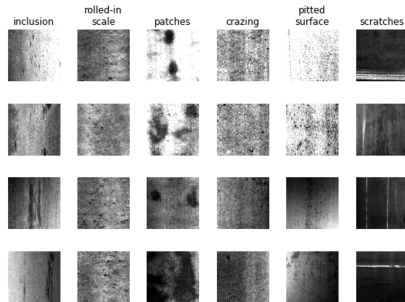


Figure: NEU Steel Surface Defects Database (NEU-SSDD)

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The Core Question

Can we achieve high-accuracy classification *without any labels*?

Our Approach: Unsupervised Learning

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 - ① **Transfer Learning:** Use a pre-trained deep neural network (VGG16) to extract meaningful features from images.

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- **No Labels Needed:** The algorithm learns to group images based on their visual features alone.
- **Core Idea:** Combine the power of two techniques:
 - 1 **Transfer Learning:** Use a pre-trained deep neural network (VGG16) to extract meaningful features from images.
 - 2 **K-Means Clustering:** A classic algorithm to partition the extracted features into distinct groups.

The Complete Unsupervised Pipeline

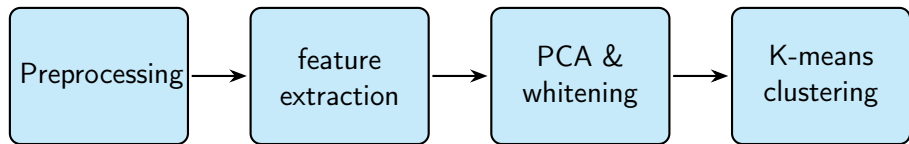


Figure: The Unsupervised Learning Pipeline.

Step 0: Image Preprocessing

Before feature extraction, two key steps prepare the images.

1. CLAHE

(Contrast Limited Adaptive Histogram Equalization)

- Normalizes brightness across images.
- Enhances local contrast to make subtle defects more visible.

2. Resizing

- Images are resized to 224x224 pixels.
- This matches the required input size for the pre-trained VGG16 network.

Step 1: Feature Extraction with Transfer Learning

Idea: Don't train a network from scratch, "borrow" a pre-trained one.

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- We feed our steel defect images through VGG16 (without its final classification layers).
- The output is a high-dimensional vector (a "fingerprint") that numerically describes the image content.
 - We primarily use the fc1 layer (4096 dimensions).

Step 2: Dimensionality Reduction with PCA

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- This retains the most important information while discarding noise.
- We also apply **whitening**, which scales each component to have unit variance, amplifying weaker (but potentially important) signals.

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- Based on the original paper, we use $k=7$ to account for six defect types plus different scratch orientations.
- **Crucially:** The algorithm is never told the true labels. It discovers the groups on its own.

Standard Pipeline Performance

Key Result

Accuracy: **99.2%**

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- The confusion matrix shows a nearly perfect diagonal, meaning the unsupervised clusters align almost perfectly with the true defect classes.

True	Cr	300	0	0	0	0	0
	In	0	298	2	0	0	0
	Pa	1	0	299	0	0	0
	PS	3	0	0	297	0	0
	RS	0	0	0	0	300	0
	Sc	0	9	0	0	0	291
		Cr	In	Pa	PS	RS	Sc
		Predicted					

Figure: Confusion Matrix (PCA=50, whitened, k=7)

Analysis 1: Is Preprocessing Necessary?

Experiment: What happens if we remove the initial Histogram Equalization (CLAHE) step?

Result: Performance Collapse

Accuracy drops from **99.2%** to **84.7%**.

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- **Conclusion:** CLAHE is a critical step. It normalizes brightness and enhances contrast, allowing the model to learn more robust features.

Confusion Matrix (NoHistEQ, PCA50 whitened, k=7)

True	Cr	300	0	0	0	0	0
	In	0	266	28	0	6	0
	PS	0	49	208	0	43	0
	Pa	0	0	0	300	0	0
	RS	0	0	0	0	300	0
	Sc	0	149	0	0	0	151
		Cr	In	PS	Pa	RS	Sc
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Figure: Confusion Matrix without CLAHE

Analysis 2: Does the VGG16 Layer Matter?

Experiment: We compared features from different VGG16 layers.

Table: Clustering Accuracy vs. VGG16 Layer Choice

Layer	Feature Space	Best Accuracy	Optimal k
fc1 (baseline)	Whitened PCA	99.2%	7
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block5_conv3	Whitened PCA	67.9%	17

Conclusion

The abstract, high-level features from the deepest **fully connected layers** (fc1, fc2) are vastly superior for this task.

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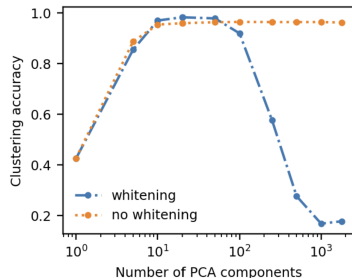
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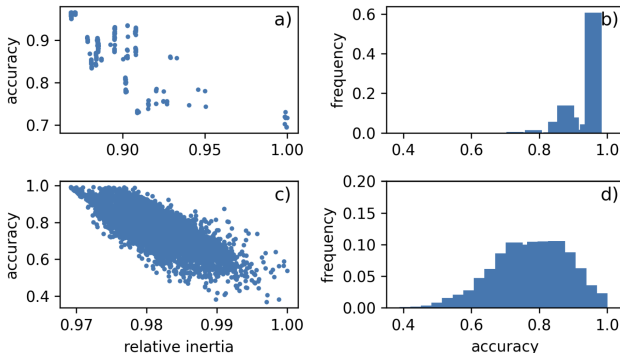


Conclusion

Whitening is crucial, but must be paired with a moderate number of components (20-50).

Analysis 4: K-Means Initialization

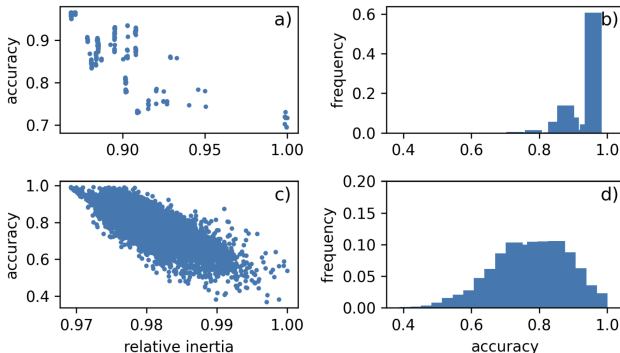
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- **Key Finding:** There is a strong negative correlation between clustering **inertia** (error) and **accuracy**.
- **Conclusion:** Whitening increases performance variability, but selecting the run with the **lowest inertia** is a very effective strategy for finding the best result (99.2%).

Analysis 5: Does the Model Generalize?

Experiment: We used 5-fold cross-validation to simulate predicting on new, unseen data.

Table: Mean 5-Fold Cross-Validation Accuracies

PCA Components	Mean Training Acc.	Mean Validation Acc.
50 (73.3 % variance)	97.2%	96.8%

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50 (73.3 % variance)	97.2%	96.8%
35 (69.1 % variance)	98.9%	99.0%

Conclusion

Yes! The pipeline can be trained as a robust predictive classifier, achieving ~99% accuracy on unseen data. A more compact 35-component representation is optimal.

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 - **Whitening** boosts performance but requires careful selection of PCA components.
 - Selecting the K-Means run with the **lowest inertia** is a reliable heuristic for success.
- Cross-validation demonstrated the model's excellent **generalizability** to unseen data.

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- **Test other clustering algorithms:** Could Gaussian Mixture Models provide better results?
- **Apply to new datasets:** Test the pipeline's robustness on different materials or defect types.
- **Semi-supervised learning:** Use the high-confidence cluster assignments as pseudo-labels to bootstrap a supervised classifier.

Thank You